North South University

Department of Electrical and Computer Engineering

CSE 215L (Programming Language II Lab)

Lab 8: Classes and Objects (II)

Objective:

- Master using class objects as parameter
- Master using an array of objects

Objects as Parameter in methods:

For so long in the methods, we have been using parameters that were of certain data types. However, now you know that class objects can also be used as Parameters.

All you need to do is to insert an object as a parameter of a method so that the object can be used inside the method for various purposes.

For example, for the Circle class, we create a method called compareGreaterRadius(Circle c), where we compare the radii of the two circles and return True if the current Circle object from where the method has been invoked has a greater radius than the radius of the Circle object parameter. The invoking of the method is given as below:

```
c1.compareGreaterRadius(c2)
```

Where c1 is the current circle object from where the method has been called and c2 is the object which was being used as a parameter.

Example:

```
class Student {

        String name;

        int age;

        // Constructor

        Student(String name, int age) {

            this.name = name;

            this.age = age;

        }
```

```java
        // Method to display student details
        void display() {
            System.out.println("Name: " + name + ", Age: " + age);
        }
    }
    class School {
        // Method that accepts a Student object as a parameter
        void printStudentDetails(Student student) {
            System.out.println("Displaying student details:");
            student.display();
        }
    }
public class Main {
        public static void main(String[] args) {
            // Create a Student object
            Student student1 = new Student("John", 20);
            // Create a School object
            School school = new School();
            // Pass the Student object to the method
            school.printStudentDetails(student1);
        }
    }
```

An array of Objects:

Do you know that it is possible to create an array of class objects, it can be declared in the same way you have been declaring an array for other data types for so long. For example, declaring an array of Circle objects is given as follows:

```
Circle [] circleArray = new Circle[10]
```

Here, we are declaring an array of Circles with size 10, meaning there can be 10 circles in the array. However, you need to initialize the circle array at first by using a for loop like this one:

```
for (int i = 0; i < circleArray.length ; i++){
circleArray[i] = new Circle();}
```

Example:

```java
class Book {
   String title;
   String author;
   // Constructor
   Book(String title, String author) {
       this.title = title;
       this.author = author;
   }
   // Method to display book details
   void display() {
       System.out.println("Title: " + title + ", Author: " + author);
   }
}

public class Library {
   public static void main(String[] args) {
       // Create an array of Book objects
       Book[] books = new Book[3];
       // Initialize the array with Book objects
       books[0] = new Book("To Kill a Mockingbird", "Harper Lee");
       books[1] = new Book("1984", "George Orwell");
       books[2] = new Book("The Great Gatsby", "F. Scott Fitzgerald");
       // Display the details of each book
       for (int i = 0; i < books.length; i++) {
           books[i].display();
       }
   }
}
```

Classwork:

1. Create a class named 'Footballer', which has the following data fields
   a. name b. team c. position d. matches e. goals f. assists

2. The footballer class should ONLY have a default constructor, having a constructor with parameters is declared ILLEGAL.
3. The footballer should have to get and set methods for name, team, and position.
4. Create methods to increase the number of matches, goals, and assists.
   a. The number of matches will increase by 1, no parameters
   b. The number of goals will increase by the provided parameters
   c. The number of assists will be increased by the provided parameters.
5. Create methods to compare the number of goals and assists with another footballer where the current footballer is being compared to another footballer. (Hint, the parameter to be used is another

a. If the current footballer has more goals and assists than the other player, then the method will print "<name of current footballer> has more goals and assists than <name of another footballer>"
b. If the current footballer has more goals but fewer or equal assists than the other player, then the method will print "<name of current footballer> has more goals but fewer or equal assists than <name of another footballer>"
c. If the current footballer has more assists but fewer or equal goals than the other player, then the method will print "<name of current footballer> has more assists but fewer or equal goals than <name of another footballer>"
d. Otherwise, print "<name of current footballer> has fewer goals and assists than <name of another footballer>"

6. Then, create two Footballers with the following information given below

	Player 1	Player 2
Name	Kylian Mbappe	Erling Haaland
Team	PSG	Manchester City
Position	Left-Winger	Center-Forward

So far, they have played 3 matches with the following statistics, given the goals and assists as user input, and should strictly follow this pattern.

	Kylian Mbappe		Erling Haaland	
	Goals	Assists	Goals	Assists
Match 1	2	0	2	1
Match 2	3	0	2	1
Match 3	2	2	2	2

Then, you will compare their goals and assist by using the comparing method you had created earlier.

Homework:

1. Carrying forward from the Footballer Class we had made in today's classwork, we will have to do the following tasks:
   a. Create an array of 15 Footballers, but you will have to initialize it.
   b. Create an additional method where you will return the number of matches.
   c. Create an additional method where you will return the average goals per match
   d. Create an additional method where you will return the average assists per match
   e. Create an additional method where you will return the average goal contributions per match.
   f. Create an additional method where you will return the goals to assist ratio
2. For those methods, use the following lines:

```
DecimalFormat df = new DecimalFormat("0.00"); // Import the DecimalFormat Class first
String roundedValue = df.format(value);
double result = Double.parseDouble(roundedValue);
```

3. Then iterate through the array and set the values of name, team, and position of each player. This time, these values should be user inputs. (No worries, all of them are strings)
4. All 15 players have played 10 matches each, thus, giving user input of their goals and assists in EACH MATCH.
5. Then print their total matches, average goals per match, average assists per match, goal contributions per match, and goals-to-assist ratio in the following manner. In the code, there won't be borders, but the output would look like this minus the border.

	Total Matches	Goals/Match	Assists/Match	Goal Contributions/Match	Goals/Assist Ratio
Player 1					
Player 2					